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Lift axle suspension systems with lift paddle

Abstract

A lift axle suspension system incorporating a lift paddle is disclosed. The lift paddle comprising a first pad, a second pad and a rigid support arm securing the first pad and the second pad in fixed relation to one another. The first pad is attached to a lift spring, the second pad is secured to the bottom surface of the main support member/leaf spring. Actuation of the lift spring causes the axle to be lifted out of contact with the ground surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a U.S. National Stage of PCT International Patent Application No. PCT/US2022/020987, filed Mar. 18, 2022, which claims priority to U.S. Provisional Application Ser. No. 63/163,539, filed Mar. 19, 2021, both of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) The present disclosure generally relates to auxiliary lift axle suspension systems for wheeled vehicles such as heavy duty trucks, commercial vehicles and trailers. More particularly, the present disclosure relates to auxiliary lift axle suspension systems having an improved mechanism for selectively lifting and lowering the wheel bearing auxiliary axle of a vehicle into and out of road engagement.

BACKGROUND OF THE INVENTION

(2) The need for certain vehicles such as heavy duty trucks, commercial vehicles and semi-trailers and the like, to have at least one (and often more than one) designated wheel bearing axle suspension system(s) capable of being raised and lowered selectively into and out of load bearing engagement with the road surface, is well-known in the art. This need usually arises in order that the vehicle be capable of legally satisfying maximum highway weight limit laws, as well as to provide an additional measure of safety when the vehicle is loaded. In this respect, such maximum weight limit laws often mandate, not just a maximum vehicle weight, but further prescribe that the required number of axles needed be spaced in such a manner so as to distribute the weight of the vehicle and its cargo over a selected length of the vehicle. Such extra axles and their attached suspension systems are often referred to as auxiliary or liftable or lift axle suspension systems.

(3) The ability to lawfully carry the maximum weight of the load (cargo) allowed by law often translates economically into maximized profit and a more economically efficient use of the vehicle.

(4) Lift axle suspension systems, however, have the potential to add to the complexity and the frequency and nature of required maintenance relative to non-liftable axle suspension systems. Ensuring safety and reliability while reducing vehicle suspension weight, complexity and purchase and maintenance costs are challenges faced when attempting to develop commercially viable, lift axle suspension systems.

(5) The present disclosure addresses shortcomings found in prior art suspension systems.

SUMMARY OF THE INVENTION

(6) There are several aspects of the present subject matter which may be embodied separately or together in the devices and systems described and claimed herein. These aspects may be employed alone or in combination with other aspects of the subject matter, and the description of these aspects together is not intended to preclude the use of these aspects separately or the claiming of such aspects separately as set forth in the claims appended hereto.

(7) In one aspect of this disclosure, a lift axle suspension assembly includes a hangar bracket, an axle and a leaf spring. The leaf spring includes a front arm and a seat. The leaf spring is configured for coupling at the location of the seat to the axle with the use of an axle coupling fastener. The front arm of the leaf spring has a top surface and a bottom surface, is of a generally solid construction, and includes an eye or is connected to an eye extension. The leaf spring is configured for pivotal mounting to the hangar bracket. The axle suspension also includes a lift bracket, a lift spring and a lift paddle arranged in an operative relationship. The first end portion of the lift bracket is secured to the hangar bracket, a second end portion of the lift bracket is connected to the first end portion of a lift spring. The paddle includes a first pad, a second pad and a first rigid support arm that connects the first pad to the second pad in fixed relation to one another. The first pad of the lift paddle attaches to the second end portion of the lift spring. The lift spring actuates and a load is transmitted from the lift spring to the lift paddle and from the lift paddle to the leaf spring causing the axle to be lifted out of contact with the ground surface.

(8) In another aspect of this disclosure, a leaf spring and lift paddle combination is provided for a

lift axle suspension. The leaf spring includes a front arm and a seat and is configured for coupling at the location of the seat to an axle. The front arm of the leaf spring has a top surface and bottom surface, is of a generally solid construction, and includes an eye or is connected to an eye extension to permit the leaf spring and lift paddle combination to be pivotally mounted to a vehicle hangar bracket. The lift paddle includes a first pad, a second pad and a first rigid support arm connected the first pad to the second pad in fixed relation to one another.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In describing the preferred examples, reference is made to the accompanying drawing figures wherein like parts have like reference numerals, and wherein:
- (2) FIG. 1 is a left side, elevational view showing both sides of a lift axle suspension system mounted to the chassis of a vehicle, incorporating a leaf spring and lift paddle, according to a first embodiment of the present disclosure;
- (3) FIG. 2 is a left side, elevational view of the left side of the lift axle suspension system of FIG. 1 with select components in phantom;
- (4) FIG. 3 is a left side, elevational view of the subject matter of FIG. 2 with the axle in a lifted position;
- (5) FIG. 4 is a left side, rear perspective view of a lift bracket from the lift axle suspension system of FIGS. 1-3;
- (6) FIG. 5 is a left side, elevational view of the bracket of FIG. 4, the right side and left side elevational views of the bracket of FIG. 4 being mirror images of one another;
- (7) FIG. 6 is a top plan view of the bracket of FIG. 4;
- (8) FIG. 7 is a bottom plan view of the bracket of FIG. 4;
- (9) FIG. 8 is a front elevational view of the bracket of FIG. 4;
- (10) FIG. 9 is a rear elevational view of the bracket of FIG. 4;
- (11) FIG. 10 is a right, rear perspective view of the lift paddle of FIGS. 1-3;
- (12) FIG. 11 is a rear, elevational view of the lift paddle of FIG. 10;
- (13) FIG. 12 is a front, right side, perspective view of the lift paddle of FIG. 10;
- (14) FIG. 13 is a front, elevational view of the lift paddle of FIG. 10;
- (15) FIG. 14 is a top plan view of the lift paddle of FIG. 10;
- (16) FIG. 15 is left side elevational view of the lift paddle of FIG. 10, the right side and left side elevational views of the subject matter of FIG. 10 being mirror images of one another;
- (17) FIG. 16 is a bottom plan view of the lift paddle of FIG. 10;
- (18) FIG. 17 is a left side, upper perspective view of a locator bracket of the embodiment shown in FIGS. 1-3;
- (19) FIG. 18 is a top plan view of the locator bracket of FIG. 17;
- (20) FIG. 19 is a bottom plan view of the locator bracket of FIG. 17;
- (21) FIG. 20 is a left side elevational view of the locator bracket of FIG. 17, the right side and left side elevational views of the locator bracket of FIG. 17 being mirror images of one another;
- (22) FIG. 21 is a rear elevational view of the locator bracket of FIG. 17;
- (23) FIG. 22 is a front elevational view of the locator bracket of FIG. 17;
- (24) FIG. 23 is a partial, top plan view of the leaf spring with selected components of the embodiment of FIGS. 1-3;
- (25) FIG. 24 is a partial bottom plan view of the subject matter of FIG. 23;
- (26) FIG. 25 is a cross-sectional, rear elevation view of the subject matter of FIG. 24, taken along lines 25-25 thereof;
- (27) FIG. 26 is a cross-sectional side view through the left side of the subject matter of FIG. 23,

taken through line **23-23** thereof;

(28) FIG. **27** is a left side, elevational view of a lift axle suspension system mounted to the chassis of a vehicle, incorporating a leaf spring and lift paddle, and with select components in phantom, according to a second embodiment of the present disclosure;

(29) FIG. **28** is a left side, elevational view of the leaf spring, eye extension and lift paddle of FIG. **27**;

(30) FIG. **29** is a left side, front perspective view of the subject matter of FIG. **28**;

(31) FIG. **30** is left side, rear perspective view of the subject matter of FIG. **29**;

(32) FIG. **31** is a top elevational view of the subject matter of FIG. **29**;

(33) FIG. **32** is a bottom elevational view of the subject matter of FIG. **29**;

(34) FIG. **33** is a rear elevational view of the subject matter of FIG. **29**;

(35) FIG. **34** is a front elevational view of the subject matter of FIG. **29**;

(36) FIG. **35** is a left side, upper perspective exploded view of a lift paddle with eye extension, a bushing, the front arm of a leaf spring and fasteners, according to a third embodiment of the present disclosure.

DETAILED DESCRIPTION

(37) The embodiments disclosed herein are for the purpose of providing a description of the present subject matter, and it is understood that the subject matter may be embodied in various other forms and combinations not shown in detail. Therefore, specific designs and features disclosed herein are not to be interpreted as limiting the subject matter as defined in the accompanying claims.

(38) A lift axle suspension system incorporating a first example lift paddle **70** of the present disclosure is indicated in general at **15** (FIG. **1**). FIG. **1** has the wheel ends and brakes removed to more clearly see the components that will be referred to or discussed herein. A longitudinally extending frame member **12** appears on each side of a vehicle chassis. A laterally extending axle **13** is disposed below the frame member **12**.

(39) The lift axle suspension system **15** incorporating a first example lift paddle **70** includes a longitudinally oriented, suspension linkage or main support member, which in this example is in the form of a leaf spring **2**. It will be appreciated that the longitudinally oriented, suspension linkage may be any suitable single leaf spring, which may be a half leaf spring which includes a front arm **6** and a seat **2** as shown in FIG. **1**, a beam spring, a full leaf spring, or a multi-leaf spring, and whether in a leading arm configuration, or in a trailing arm configuration as shown in FIG. **1**. The leaf spring **2** in the lift axle suspension system **15** shown in FIG. **1** extends over the axle **13** in an overslung arrangement. In other suspension systems, the suspension linkage may extend over, under, through or around the axle **13**, without departing from the scope of the present disclosure.

(40) The axle **13** is connected or secured to the leaf spring **2** through the use of an axle coupling assembly, which in this example, rigidly secures the axle **13** to the leaf spring **2** at the location of the seat **4** in the manner discussed below. The leaf spring **13** has a first end that includes a cylindrical eye portion or eye **8** movably connected or pivotally connected to the frame hanger bracket **11** with materials and techniques known to those skilled in the art, which may include for example, the use of a bushing assembly **8a** and a pivot bolt **9**, a thread nut or connector **9a**, and an end plate or end cap **9b**, as appropriate. The axle **13** and the leaf spring **2** pivot about a rotational axis. During normal use, the travel path or movement of the axle **13** and the leaf spring **2** generally follows an arc defined in substantial part by the main support member's pivotal connection to the vehicle frame via hanger bracket **11**.

(41) The axle coupling assembly operatively couples the axle **13** to the leaf spring **2**. From the top downward beginning with the top mounting pad, the axle coupling assembly includes a top mounting pad or first mounting pad **21**, also known as a first guide plate or clamp plate; a caster wedge or intermediate pad **22** disposed between the seat **4** of the leaf spring **2** and the axle **13**; and a bottom mounting pad or second mounting pad **24**, also known as a second guide plate or clamp plate which includes a reward extension **26** providing platform for mounting an air spring **30** which

is also mounted at its top end to the vehicle frame member **12**. A pair of U-shaped fastener assemblies **25**, **27**, which in the first example embodiment are shown as first and second U bolts and connectors are used to clamp together the aforementioned components in fixed relation to another. A shock absorber **34** is pivotally connected at its upper end to a bracket mounted on the vehicle frame member **12** and attached at its bottom end to a rearward extension **24** of the bottom mounting pad **23**. Although the present disclosure employs an air spring **30**, it is not limited to such a configuration, and may be deployed with other suspension types, such for example, the suspension type illustrated in U.S. Pat. No. 9,050,873 which employs a straight monoleaf spring design with shock absorber **34**, with and without an air spring, as well as other suspension types referred to or disclosed herein without departing from the scope of the present disclosure.

(42) A leaf spring retainer **32**, is constructed as a band to wrap around and capture the leaf spring **2**, in the event the spring is fractured during vehicle operation. In FIG. **2**, the leaf spring retainer is secured at one end between the top pad **21** and the seat **4** of the leaf spring **2**. The leaf spring retainer extends forward and around the eye **8** to be secured between the bottom surface of the front end portion of the front arm **6** of the leaf spring **2** and the top pad or the second pad **76** of the paddle **70**.

(43) A lift bracket **50** (FIGS. **4-9**) to which a first end portion **62** of a lift spring **60** is secured at a first end portion or upper portion to the frame hangar bracket **11**. The lift bracket **50** serves as an extension of the hangar bracket **12** and provides a generally stationary platform in relation to the lift spring **60** and lift paddle **70** for mounting the lift spring **60**. In this example, the lift bracket **50** includes a body **52** having a first end portion or upper end portion configured for attachment to the hangar bracket. A second end or lower end portion presents a disk or pad **54** for mounting a first end portion **62** of the lift spring **60**. A second end or the movable end **64** of the mounted lift spring **60** is received by or secured to the first example lift paddle **70** as further discussed below.

(44) FIG. **2** shows one side of the lift axle suspension system of FIG. **1** with the hangar bracket in phantom to better show the first example lift paddle **70** secured to the front arm **6** of the leaf spring **2** rearward of and near the eye **8**. A similar system would be provided on the opposite side of the vehicle.

(45) FIG. **3** shows the subject matter of FIG. **2** in a lift position which would lift the wheels (not shown) of the axle out of contact with the ground surface. In lift mode, the lift spring is actuated, causing the lift spring to transmit a load to the lift paddle, which lifts the leaf spring and axle in a manner that causes the axle to no longer be in a position of engagement, through its wheels (not shown), with the ground surface. An air spring (if present) may release some or all of the downward pressure that it exerts when the axle is in a position of engagement with the ground surface.

(46) In the first example of the present disclosure, a paddle **70**, shown separately in FIGS. **10-16**, includes a first pad **72** that connects to a second pad **76**. The first pad **72** and the second pad **76** each present an engagement surface on the exterior of the lift paddle **70** for their attachment or fixed securement to, respectively, the movable end **64** of the lift spring **60** and the bottom surface of the leaf spring. In the example of FIGS. **1-3**, the generally planar, exterior engagement surfaces of the first pad **72** and the second pad **76** are generally perpendicular or at slightly obtuse angle to one another. However, the size, spacing, orientation and geometry of the first pad **72** and the second pad **76** and their respective engagement surfaces may vary in accordance with the size, spacing, orientation and geometry of the suspension components to which they attach without departing from the scope of the present disclosure. Corresponding bores (**76a**, **6a**) defined in the second pad, and in front arm **6** of the leaf spring **2** are positioned in registration and receive a suitable fastener **77**.

(47) The first pad **72** and the second pad **76** connect or join on the exterior of the lift paddle **70**, directly, or indirectly via a generally planar, intermediate pad, wall or plate **74**, that forms a bridge, such as, for example, as shown in FIG. **12**. The first pad **72**, the intermediate pad **74**, and the

second pad **76**, together forms a continuous, exterior wall that extends from the first pad's attachment to the movable end **64** of the lift spring **60** to the second pad's fixed securement to the leaf spring **2**. A vertical cross-section or side elevational view of this part of the first paddle, as shown in FIG. **15**, may be defined or characterized by three segments, each having a different angle of orientation. As shown in FIG. **15**, the generally vertical first segment associated with the first pad and the generally horizontal last segment associated with the second pad connect via the segment associated with the intermediate pad, which in the example shown in FIG. **15**, has an orientation distinct from the first and third segments.

(48) The lift paddle **70** of this first example also includes first and second rigid, support arms **80**, **84**. The first and second rigid, support arms **80**, **84** are generally planar, generally parallel to one another and spaced apart in the lateral dimension of the lift paddle **70**, being positioned respectively on the inboard and outboard sides of the lift paddle **70**. In the first example, first and second rigid support arms have the same shape and size, however, in another aspect of the present disclosure, they may also vary without departing from the scope of the present disclosure. The front and top surfaces or edges of the first and second rigid, support arms **80**, **84** are joined or are otherwise secured, by weldment as shown in FIGS. **10** and **15** or by other suitable fastening means to the first pad **72** and to the second pad **76** and to any intermediate pad **74** or bridge by which the first pad **72** and the second pad **76** connect or join. In the first example, the front and top surfaces or edges of the first and second rigid support arms continuously follow the interior geometry of the paddle **70** defined by the first pad **72**, the intermediate pad **74** and the second pad **76** (see, e.g., FIGS. **10**, **12** and **15**) as the first and second rigid, support arms traverse the paddle **70** from the top downward. This construction helps to efficiently handle and distribute stress loads and to minimize stress concentrations.

(49) From the top downward, the rear edges or rear surfaces of the first and second rigid, support arms **80**, **84** descend from the rear edge of the bottom surface of the second pad, generally straight down. They then make an arcuate turn until the tangent of the rear edges or rear surfaces are generally horizontal which is approximately the elevation of the bottom edge of the intermediate pad **74**. The rear edges or rear surfaces of the first and second rigid support arms **80**, **84** then makes another turn in the opposite direction to continue their descent as they create a point or an area of inflection. At or slightly above the top of the bottom pad **72**, the rear edges or rear surfaces of the first and second rigid, support arms **80**, **84** orient in a generally straight line that forms in the lower half of the paddle **70** a wedge shape that converges with the interior surface of the first pad at the bottom of the paddle **70**.

(50) When the first and the second rigid, support arms **80**, **84** are secured by weldment or other suitable fastening means to the first pad **72** having a first orientation, the second pad **76** have a second orientation, and to an intermediate pad **74** or bridge, having a third orientation that is between the first orientation and the second orientation, the example lift paddle **70** achieves the strength to react the dominant compressive and tensile loads required to lift and to maintain the axle **13** out of contact with the ground, while retaining the compliance necessary to accommodate forces that attempt to twist the lift paddle **70** which have the potential of damaging the lift spring **60** or other attached componentry if not properly managed. Such forces might be experienced by the leaf spring when it twists or rotates, such as for example, during roll motion or cross axle articulation. During these events, one side of an axle moves upward while the other side of the axle moves downwards, resulting in twisting or rotation of the leaf spring about its longitudinal axis. Such forces may be greater in a trailing leaf spring suspension than in a trailing stiff beam suspension.

(51) A locator bracket **40**, shown separately in FIGS. **17-22**, is fixedly and reversibly secured, with the use of a threaded fastener **77**, to the first example lift paddle **70**, to facilitate the longitudinal alignment of the leaf spring and the first bracket during installation and to maintain this alignment during vehicle operation. The locator bracket **40** includes a generally horizontal base **42** and a

generally vertical arm portion. The base **42** generally resembles a T shape and includes a first portion that includes an elongated stem or backbone. The first portion of the base **42** is received in a slot formed by the parallel, spaced apart, first and second rigid support arms **80, 84** on the underside of the top pad **76**. The lateral edges of the first portion of the base **42** extend in parallel and then taper **43** before they flare laterally to form a shoulder portion **44** of the base **42** that includes first and second lateral end portions, which then extend vertically while spaced apart to form, respectively, a first arm **46** and a second arm **48**, each of which may abut a lateral surface at the neutral axis **3** of the leaf spring **2** received between the first arm **46** and the second arm **48** (FIGS. **24, 25**). Respective bores (**6a, 76a, 42a**) defined in each of the forward end portion of the front arm **6** of the leaf spring **2** (rearward of the eye **8**), the top pad **76** of the lift paddle **7**, and the base **42** of the locator bracket **40** are brought into registration by relative positioning of these structures to receive a fastener **77** and form a leaf spring, lift paddle **170** assembly of the present disclosure (FIG. **26**). The front end of the leaf spring retainer **32** also is secured by fastener **77** in leaf spring retainer bore **32a** (FIG. **33**).

(52) It should be noted that while the lift paddle **70** of the first example is illustrated with the use of a leaf spring **2** that includes an eye **8**, it also would be operable with a leaf spring wherein the front arm is connected to an eye extension.

(53) A second example of the present disclosure is shown in FIGS. **27-34** and includes many of the same components previously described in the first embodiment. Accordingly, the same reference numerals and descriptions above with respect to the first example apply as to those same components.

(54) A lift axle suspension system incorporating the second example lift paddle **170** is indicated in general at **115** (FIG. **27**). This second example lift paddle **170** is suitable for use with a leaf spring manufactured without an eye or other distal end configuration intended to facilitate pivotal connection of the leaf spring with a vehicle frame **12** or vehicle frame hanger **11**. The leaf spring **102** of this second example includes a front arm **106** and a seat **104** and is connected or secured at the front end of the front arm **106** to an eye extension **108**.

(55) The lift paddle **170** of this second example includes a first pad **172** and a second pad **176**, which are spaced apart and integrally formed in, or secured to, in fixed relation to one another in a frame, which also includes a first arm or front arm **180**, a second arm or rear arm **184**, and a third arm or interior arm **183**. The first pad **172** is positioned to present a front or an exterior or an engagement surface for attachment to a second end or movable end **64** of a lift spring **60**. The first pad **172** is generally vertically oriented and positioned in a lower and forward portion of the lift paddle **170**, and is integrally formed with or fixedly secured with the use of other fasteners to a first arm **180** configured as an elongated rail or frame member traversing the first pad **172** on its rear surface. The second pad **176** is positioned in an upper portion of the lift paddle **170** and presents an upwardly facing engagement surface for engaging a bottom surface of the leaf spring **102**.

(56) The first arm or front arm **180** of the frame has a lower or first end portion integrally formed with or fixedly secured with the use of other fasteners to the first pad **172**, from which the first arm **180** extends in a generally vertical direction. As the first arm **180** reaches the top of the lift paddle **170**, it forms an arc in the upper front section of the lift paddle **170** after which an upper or second end portion of the first arm **180** is integrally formed with or secured with the use of other fasteners to the forward end portion or the rear surface of the top pad or second pad **176** of lift paddle **170**.

(57) A second arm or rear arm **184** of the frame has a lower or first end portion integrally formed with or secured with the use of other fasteners to the bottom rear portion of the first pad **172**, from which the second arm **184** extends diagonally, in a vertical and horizontal direction away from the first pad **172**. At or slightly above the top edge of the engagement surface of the first pad **172**, the second arm **184** changes its orientation to ascend primarily in a vertical direction that is generally perpendicular to the leaf spring **102**, increasing in size in the lateral dimension of the lift paddle **170**, until it reaches the top of the lift paddle **170** where it then turns in the forward direction to

present a top or second end portion that is integrally formed with or secured with the use of other fasteners to the rear end portion or the rear surface of the top pad or second pad **176** of lift paddle **170**. The increase in size of the second arm **184** in the lateral dimension of the lift paddle **170** assist the load capacity of the lift paddle **170** and, when fixedly secured to an eye extension **108** as described and illustrated herein, assists the lift paddle **170** spring combination in resisting in forces that attempt to twist or rotate the leaf spring about its longitudinal axis.

(58) A third arm or interior arm **183** of the frame extends across the interior of the frame of the second example lift paddle **170**. A first end or forward end of the third arm is integrally formed with or secured with the use of other fasteners to the first arm or the front arm **180** at or slightly above the first pad **172**. A second end or rear end of the third arm is integrally formed with or secured with the use of other fasteners to the second arm or the rear arm **184**. When configured in this manner, the third arm or the interior arm **183** helps to define an upper section and a lower section of the lift paddle **170**: the upper section having four sides that define a longitudinally and vertically disposed window **188**, and the lower section generally defining a longitudinally and vertically disposed triangle or wedge **186** that is largest in the mid-section of the paddle **170** and converges with rear surface of the first pad **170** at the base of the first pad **172**. A wall fills the interior of the lower section for added strength. The absence of a wall in the upper section reduces the weight of the paddle **170**. It will be appreciated that the upper section and the lower section of the lift paddle **170** may be varied in size and shape without departing from the present disclosure.

(59) The eye extension **108** of the second embodiments includes an eye in the form of a cylinder or generally cylindrical portion **108a** configured to receive a bushing assembly **108b**. The exterior surface of the cylinder **108a** includes a first mounting provision **109** extending generally downward and a second mounting provision **110** extending generally rearward, which together permit the eye extension **108** to be attached or secured to each of the lift paddle **170** and the front end portion of the front arm **106** of a leaf spring **102**, and when secured together, allows this assembly to be pivotally mounted to a vehicle hangar bracket **11**.

(60) In this second example, the first mounting provision **109** and the second mounting provision **110** are extensions positioned and spaced apart on the exterior surface of the cylindrical portion **108a** of the eye extension **108**.

(61) The first mounting provision **109** of the eye extension **108** is located between the five and six o'clock position of the generally cylindrical portion **108a** and is configured to abut a first mounting section **182** at the first arm **180** of the paddle **170** between the top of the first pad **172** and the top of the lift paddle **170**. As shown in FIG. **28**, respective bores defined in the first mounting provision **109** of the eye extension **108** and the first mounting section **182** of the first arm **180** of the paddle **170** are placed in registration by the relative positioning of the eye extension **108** and the lift paddle **170**, to receive a fastener **181** to assist in fixedly securing the eye extension **108** and the lift paddle **170** together.

(62) The second mounting provision of the eye extension **108** is located approximately eighty (80) to one hundred (100) degrees counterclockwise from the first mounting provision. degrees. The front end portion of the front arm **106** of the leaf spring **102** is positioned between the second mounting provision **110** of the eye extension **108** and the second pad **176** of the lift paddle **170** to place respective bores in each of these three structures in registration. Fasteners (**177a**, **177b**, **177c**, **177d**) received in the respective bores (**110a**, **110b**, **110d**; **106a**, **106b**, **106c**, **106d**; **176a**, **176b**, **176c**, **176d**) are tightened to assist in securing these three structures in fixed securement relative to one another. The front end of the leaf spring retainer **132** also is secured by fasteners (**177c**, **177d**) in leaf spring retainer bores **132c**, **132d** (FIG. **33**).

(63) A third embodiment of the present disclosure is shown in FIG. **38**, and includes many of the same components previously described in the first embodiment. Accordingly, the same reference numerals and descriptions above with respect to the first example apply as to those same components. In a third embodiment of the present disclosure (FIG. **35**), the example lift paddle **270**

and eye extension **208** including a bushing assembly **208b** are integrally formed in one piece construction.

(64) Each of the structures of the foregoing examples may be constructed from various materials including steel, ductile iron, aluminum, alloys or other suitable substantially rigid materials. It will also be appreciated that the pieces may be manufactured by any suitable method, such as by casting, forging, machining or the like. The same material and/or method of manufacturing need not be used for each piece. Manufacturing of knuckle componentry also may include machining, for example, to drill and/or tap bores or to flatten engagement surfaces, as further discussed herein.

(65) It will be appreciated that throughout this disclosure, with respect to the descriptions of all of the examples provided, terms such as upper, lower, top, bottom, left, right, front and rear, all are relative terms and are not intended to be limiting because they depend on the context in which they are used and the relative position of the device at the time.

(66) It will be understood that the embodiments described above are illustrative of some of the applications of the principles of the present subject matter. Numerous modifications may be made by those skilled in the art without departing from the spirit and scope of the claimed subject matter, including combinations of features that are individually disclosed or claimed herein. For these reasons, the scope of this disclosure is not limited to the above description but is as set forth in the following claims, and it is understood that claims may be directed to the features hereof, including as combinations of features that are individually disclosed or claimed herein.

Claims

1. A lift axle suspension system comprising, hanger bracket configured for attachment to a vehicle frame; an axle; a leaf spring, comprising a front arm and a seat, the front arm extending forwardly from the seat, the front arm having a top surface and a bottom surface and being of a generally solid construction, the leaf spring configured for coupling at the location of the seat to the axle with the use of an axle coupling fastener, the front arm of the leaf spring including an eye or connected to an eye extension, the leaf spring configured for pivotally mounting to the hanger bracket; a lift bracket and a lift spring, the first end portion of the lift bracket secured to the hanger bracket, the second end portion the lift bracket connected to the first end portion of the lift spring; a lift paddle comprising a first pad, a second pad, and a first rigid support arm, the first pad providing an engagement surface for attaching the first pad to the second end portion of the lift spring, the second pad providing an engagement surface for securing the second pad to the bottom surface of the front arm of the leaf spring, and the first rigid support arm connecting the first pad and the second pad to support the fixed position of the first pad and the second pad in relation to one another; the front arm of the leaf spring defining a bore, the second pad of the lift paddle defining a bore, the bore of the front arm of the leaf spring extending through the leaf spring and positioned in registration with the bore of the second pad, the bores of the front arm of the leaf spring and the second pad receiving a fastener securing the front arm of the leaf spring to the second pad of the lift paddle; wherein the lift bracket, the lift spring and the lift paddle are configured for arrangement in an operative relationship with one another wherein by actuation of the lift spring, a load is transmitted from the lift spring to the lift paddle and from the lift paddle to the leaf spring causing the axle to be lifted out of contact with the ground surface.

2. The lift axle suspension of claim 1 wherein the lift spring comprises an air spring.

3. The lift axle suspension of claim 1 wherein the lift spring is positioned below the leaf spring.

4. The lift axle suspension of claim 1, wherein the lift spring is configured to expand and contract along a central operational axis that is parallel to or that does not intersect, the front arm of the leaf spring.

5. The lift axle suspension of claim 1, wherein the paddle further comprising a second rigid, support arm connecting the first pad and the second pad to further support the fixed position of the

first pad and the second pad in relation to one another.

6. The lift axle suspension of claim 5, wherein the first pad and at least one of the first rigid support arm and the second rigid support arm forms a wedge that converges at the bottom of the lift paddle.

7. The lift axle suspension of claim 5, wherein the first rigid support arm and the second rigid support arm are positioned spaced apart from one another on opposite sides of the lift paddle.

8. The lift axle suspension of claim 1 wherein the front arm of the leaf spring includes the eye which is integrally formed with the leaf spring in one piece construction.

9. The lift axle suspension of claim 5 wherein the first rigid support arm and the second rigid support arm are similar in shape and in size and are positioned, respectively, on the inboard side and the outboard side of the lift paddle to form respectively, an inboard face and an outboard face of the lift paddle.

10. The lift axle suspension of claim 5 where the first rigid support arm and the second rigid support arm are plates.

11. The lift axle suspension of claim 5 wherein the first pad and the second pad further connect to one another through an intermediate pad to form a continuous wall, and wherein the first pad, the intermediate pad, and the second pad each have a different orientation.

12. The lift axle suspension of claim 11 wherein the first pad, the intermediate pad, and the second pad are formed from the same piece of material having a first bend line and a second bend line, the first bend line being generally parallel to the second bend line, the first bend line representing a boundary line between the first pad and the intermediate pad, the second bend line representing a boundary line between the intermediate pad and the second pad.

13. The lift axle suspension of claim 1, the lift paddle including a locator bracket configured to aid lateral alignment of the front arm of the leaf spring and the lift paddle during installation and during vehicle operation, the locator bracket comprising a base and an arm portion, the base having a width and a length that provides a first longitudinal end portion and a second longitudinal end portion, the top surface of the base being generally planar, the arm portion having a first arm and a second arm positioned across from one another on opposite sides of the second longitudinal end portion and extending in a generally vertical direction in relation to the base portion, wherein the base of the first longitudinal end portion of the locator bracket is secured in fixed relation to the second pad of the lift paddle at a location that positions the first arm and the second arm of the locator bracket to extend above the top surface of the second pad and to receive the front arm of the leaf spring between the first arm and the second arm of the locator bracket.

14. The lift axle suspension of claim 13 wherein the base of the locator bracket is positioned below and in contact with the bottom surface of the lift paddle, placing the first arm and the second arm of the locator bracket rearward of the second pad, and wherein front arm of the leaf spring, the second pad of the lift paddle and the locator bracket are secured fixed together with at least one fastener.

15. The lift axle suspension of claim 5 wherein the first rigid support arm and the second rigid support arm are dissimilar in shape and in size and are spaced apart in the longitudinal dimension of the lift paddle, the first rigid support arm positioned in the front of the lift paddle, the second rigid support arm positioned in the rear of the lift paddle.

16. The lift axle suspension of claim 5 wherein the lift paddle including a longitudinally and vertically disposed window at least partially defined by at least a portion of the first rigid support arm positioned in the front of the lift paddle and at least a portion of the second rigid support arm positioned in the rear of the lift paddle.

17. The lift axle suspension of claim 5 wherein the at least a portion of the second rigid support arm increases in size in the lateral dimension as the portion extends in the vertical direction.

18. The lift axle suspension of claim 5, the front arm of the leaf spring connected to the eye extension, the eye extension including a cylindrical portion and a first mounting provision and a second mounting provision spaced apart and positioned on an exterior surface of the cylindrical portion, the first rigid support arm of the lift paddle including a mounting section positioned in the

front of the lift paddle, the first mounting provision of the eye extension fixedly secured to the mounting section, the front arm of the leaf spring having a front end portion positioned between the second mounting provision of the eye extension and the top surface of the second pad of the lift paddle, the second mounting provision of the eye extension, the front end portion of the front arm of the leaf spring, and the second pad of the lift paddle fixedly and reversibly secured together with the fastener.

19. The lift axle suspension of claim 18 wherein the first mounting provision of the eye extension is integrally formed in one piece construction with the mounting section of the first rigid support arm of the lift paddle.

20. The lift axle suspension system of claim 1 wherein the fastener securing the second pad of the lift paddle to the leaf spring is the sole fastener securing the lift paddle to the leaf spring.

21. A leaf spring and lift paddle combination for a lift axle suspension, the leaf spring and lift paddle combination comprising a leaf spring comprising a front arm and a seat, the front arm extending forwardly from the seat, the front arm having a top surface and a bottom surface and being of a generally solid construction, the leaf spring configured for coupling at the location of the seat to an axle, the front arm of the leaf spring including an eye or connected to an eye extension to permit the leaf spring and lift paddle combination to be pivotally mounted to a vehicle hangar bracket; a lift paddle comprising a first pad, a second pad, and a first rigid support arm, the first pad of the lift paddle providing an engagement surface to attach the first pad to a movable end of a mounted lift spring, the second pad of the lift paddle providing an engagement surface for securing the second pad to the bottom surface of the front arm of the leaf spring, and the first rigid support arm connecting the first pad and the second pad to support the fixed position of the first pad and the second pad in relation to one another, the front arm of the leaf spring defining a bore, the second pad of the lift paddle defining a bore, the bore of the front arm of the leaf spring extending through the leaf spring and positioned in registration with the bore of the second pad, the bores of the front arm of the leaf spring and the second pad receiving a fastener securing the front arm of the leaf spring to the second pad of the lift paddle.
